

RPG-Massey Hypothetic Research Plan for formal validation - rev1

As a **research roadmap**, it is surprisingly good. As a **realistic Lean 4 project proposal**, it is simultaneously ambitious and somewhat optimistic about the current state of formalized algebraic topology.

My assessment would be:

Phase	Mathematical validity	Lean feasibility
Phase 1	A	A–
Phase 2	A	C
Phase 3	A–	B–
Phase 4	A	D

Overall: **conceptually excellent, practically a 10-20 year formalization program unless a large team participates.**

Strongest aspect

The proposal correctly identifies the logical chain connecting the two essays:

$\mathbb{RP}^n \rightarrow \gamma^1 \rightarrow w(\mathbb{RP}^n) \rightarrow \text{dual SW classes} \rightarrow \text{immersion obstructions} \rightarrow \text{Massey} \rightarrow \text{Cohen}.$

That is exactly the mathematical route by which projective geometry enters immersion theory.

In particular, the observation

products of projective spaces are the extremal examples
captures the historical role of projective spaces very well.

Phase 1

This is realistic.

$\alpha(n)$

Completely feasible.

One could define

```
def alpha (n : ℕ) : ℕ :=  
  Nat.popcount n
```

rather than building it from binary digits manually.

Mathlib already contains substantial binary arithmetic infrastructure.

Projective space

Mostly feasible.

However:

Lean's mathlib4 already has the Projectivization of a vector space
is true,
but

formally endow it with a SmoothManifold structure
is a nontrivial project.

The manifold structure on projective space is not currently one of the standard examples that enjoys complete API support comparable to Euclidean spaces or spheres.

Still, this is definitely within reach.

Immersions

Also feasible.

Mathlib's manifold framework already knows about derivatives, tangent bundles, and smooth maps.

Phase 2

This is where difficulty explodes.

The proposal understates the challenge.

Tautological bundle

Reasonable.

This is difficult but manageable.

Projective space and Grassmannians are classical geometric constructions.

Singular cohomology

The statement

requires setting up singular cohomology
is actually the understatement of the document.

Today, the main bottleneck is not Stiefel-Whitney classes.

It is the enormous amount of homotopy-theoretic infrastructure underneath them.

You would need:

- singular chains,
- cochains,
- cup products,
- Eilenberg-Zilber machinery,
- mod-2 coefficients,

- naturality,
- bundle classification.

This is already a major standalone project.

Stiefel-Whitney classes

This is arguably a research-level formalization project by itself.

The proposal makes it sound like:

define SW classes and prove Whitney sum formula.

In practice this could easily require years of work.

Phase 3

This section is mathematically sensible.

However:

```
theorem massey_obstruction ...
```

is not actually how one would formalize the theorem.

The type

```
SmoothManifold ℝ n
```

does not correspond to “all n-manifolds.”

One would need a much richer manifold structure.

More importantly, the theorem would probably be expressed in terms of:

- stable tangent bundles,
- stable normal bundles,
- characteristic classes.

The pseudocode is conceptually correct but not close to actual Lean.

$\mathbb{RP}^{2^k}$ obstruction

This is probably the best intermediate target in the entire proposal.

Once SW classes exist, proving

$$w(\mathbb{RP}^{2^k}) = (1 + a)^{2^k + 1}$$

and extracting nonvanishing classes is a very natural formalization milestone.

This is exactly the kind of theorem one would expect before touching Massey or Cohen.

Phase 4

This is where the proposal becomes genuinely visionary.

Cobordism

Not currently available.

One essentially needs:

$$MO$$

and Thom spectra.

That alone is an enormous body of formalized stable homotopy theory.

Surgery theory

Even harder.

Modern surgery theory relies on:

- homotopy groups,
- normal maps,
- obstruction groups,
- Wall's L-theory.

Virtually none of this exists in a mature formalized form.

BP-theory

This is the most optimistic sentence in the document:

Defining the spectrum BP and formalizing BP-homology of BO.

That is not a "next step."

That is a research program.

Even formalizing spectra in a form robust enough for Brown-Peterson homology is a substantial project.

One thing I would change

I would split the project into three separate goals.

Goal A

Formalize

$$\mathbb{R}P^n$$

and its topology.

Deliverables:

- tautological bundle,
- cohomology ring,
- SW classes,
- proof

$$w(\mathbb{RP}^n) = (1 + a)^{n+1}.$$

This alone would already be a notable contribution.

Goal B

Formalize immersion obstructions.

Deliverables:

- stable normal bundle,
- dual SW classes,
- nonimmersion theorems for projective spaces.

At this point one can already recover many classical results.

Goal C

Formalize Massey and eventually Cohen.

This should be viewed as a separate long-term project.

The jump from

$$w(\mathbb{RP}^n) = (1 + a)^{n+1}$$

to

$$\text{every manifold immerses in } \mathbb{R}^{2n-\alpha(n)}$$

is vastly larger than the roadmap suggests.

Consistency with your projective-geometry essay

Very high.

In fact, if your essay were the mathematical narrative and the roadmap were an implementation plan, they fit together naturally:

1. Real projective geometry introduces $\mathbb{RP}^n$.
2. The topology chapter introduces the tautological bundle and Stiefel-Whitney classes.
3. Massey's theorem explains why $\alpha(n)$ appears.
4. Cohen's theorem explains the immersion bound quoted in §6.3.
5. Lean formalization provides machine verification of that chain.

So as a **research direction statement**, I would rate it around **8.5/10**.

As a **practical estimate of the amount of formalization work required**, I would revise the expectations upward significantly: proving Cohen's theorem in Lean is not merely a large theorem formalization, but would likely require building a substantial fraction of modern characteristic-class theory, cobordism theory, and stable homotopy theory first.